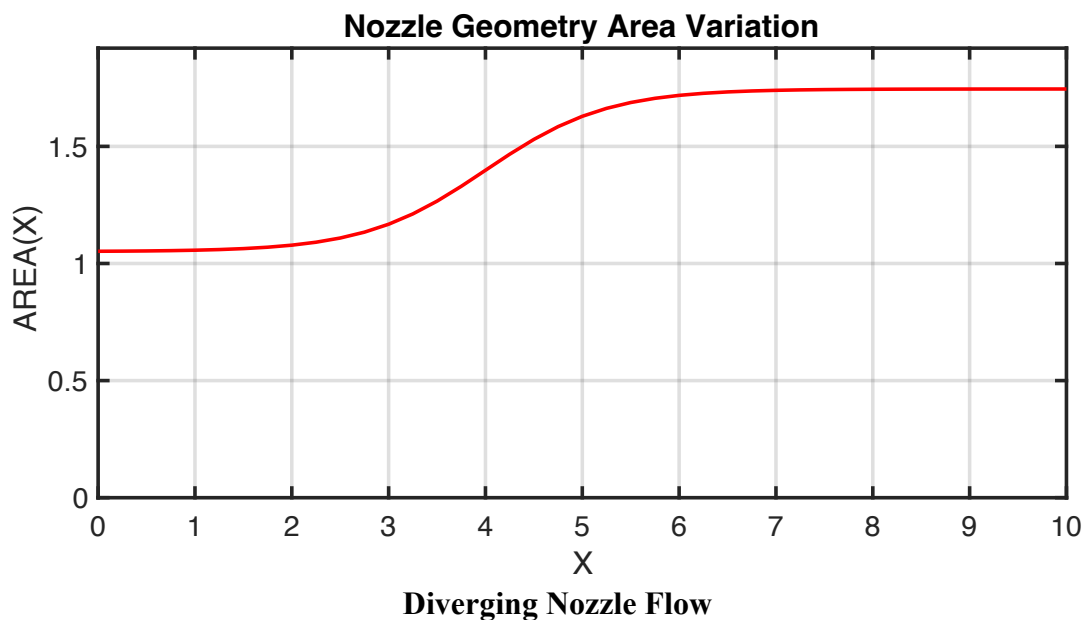


CFD Lab 2026 Assignment

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Nozzle Flow: Transonic Flow in A Quasi-1D Nozzle

Your boss noticed a set of MATLAB routines and programs for examining transonic flow in nozzles and has provided them for you (along with a couple of PDF files). She is curious about its purported capabilities to do space marching and unsteady flow in nozzles. She would like you to clean-up the code a bit and demonstrate its usefulness and applicability to several problems of interest. Please note that the code uses non-dimensionalized variables.

**CFD:**

Introductory lecture on CFD were given in class and are available as a handout, and therefore won't be discussed here.

Steady Flow:

For steady flow the problem can be handled using space marching and shock fitting as described in a handout (Space_Marching.pdf) and in class and won't be discussed here.

Unsteady Flow:

For unsteady flow we need to consider the governing equation, space discretization, initial and boundary conditions, and the interior operator (if you are interested in more details they are in Upwind_Hirsch.pdf; unfortunately details are beyond one or two lectures).

Governing Equation. If the area change is gradual in space and time and viscous effects are small then the unsteady flow in the nozzle can be approximated in a quasi one dimensional manner (through the inclusion of a source term) by the following form of the Euler equations:

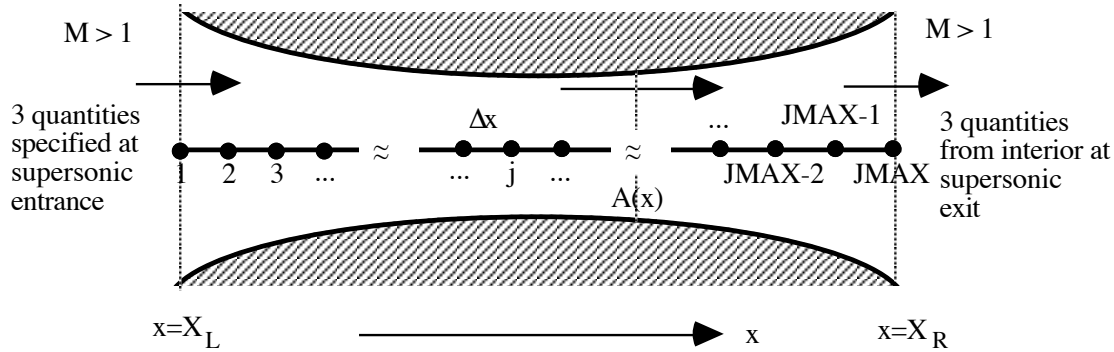
$$\frac{\partial \hat{Q}}{\partial t} + \frac{\partial \hat{F}}{\partial x} = \hat{V}_s$$

where

$$\hat{Q} = \begin{pmatrix} \rho A \\ \rho u A \\ e A \end{pmatrix}, \quad \hat{F} = \begin{pmatrix} \rho u A \\ (\rho u^2 + p) A \\ (e + p) u A \end{pmatrix}, \quad \hat{V}_s = \begin{pmatrix} 0 \\ p \frac{\partial A}{\partial x} \\ 0 \end{pmatrix}$$

$$p = (\gamma - 1)(e - 0.5 \rho u^2)$$

Discretization, Boundary Conditions, Initial Condition. The governing equation is discretized in space and time. Equal spacing along the nozzle is used:



Transonic Nozzle Flow Discretization and Boundary Conditions

The cross sectional area along the nozzle is given by (for a diverging nozzle):

$$A(x, t) = 1.398 + 0.347 \tanh(0.8(x - 4.)) \quad 0 \leq x \leq 10$$

For the diverging nozzle problem the three physical boundary conditions at the entrance are $M_L = 1.50$, $p_L = 0.25$, and $\rho_L = 0.50$ (assume $\gamma = 1.4$ everywhere) and they remain fixed for all time. There are no numerical boundary conditions at the entrance. Compatibility relations are used to update the quantities at the exit characteristically. Assume that the nominal case is for a shock that is located in the nozzle such that the exit pressure is atmospheric ($p_R = 0.7143$). The initial condition throughout the nozzle can be set using space marching techniques.

Interior Operator. A Flux Vector splitting technique is used to calculate the flux contributions. The explicit flux vector split method interior operator can be written as:

$$\Delta \hat{Q}_j^n = -\frac{\Delta t}{\Delta x} (\hat{F}_j^{+n} - \hat{F}_{j-1}^{+n} + \hat{F}_{j+1}^{-n} - \hat{F}_j^{-n}) + \frac{\Delta t}{\Delta x} \begin{pmatrix} 0 \\ p_j^n (A_{j+1/2}^n - A_{j-1/2}^n) \\ 0 \end{pmatrix}$$

$$RESID_j^n = \Delta \hat{Q}_j^n$$

$$\hat{Q}_j^{n+1} = RESID_j^n + \hat{Q}_j^n$$

This is implemented in the MATLAB code and applied at each time step. The time step size, Δt , must be small enough to not produce numerical instabilities.

Documentation (remember to follow any directions from Dr. Silbaugh):

A) Description

Include a short description of the problem and the theory behind quasi-1d nozzle flow. Briefly describe the computer code implementation, as well as mention what type of computer was used. Please modify *sample.m* and *quasild.m* to produce the following requested results (this should be relatively simple). As part of this you may need to write some of your own MATLAB code or clean-up the existing code (especially making sure that all plots are clearly labeled with appropriate titles and axis labels).

B) Nozzle Geometry

Include one plot that shows the nozzle geometry examined ($\pm r(x)$ versus x). Since we are considering a round jet, $A = \pi r^2$. Clearly label the plot and refer to it in the text (1st plot). NOTE that you might need to modify the code slightly.

C) Space Marching

Plot the pressure versus x for a fixed shock location, $x_{sh} = 4.0$, and for a fixed total number of points, $j_{max} = 81$ ($i_{refine}=2$). Plot the results from using just the predictor in space marching and from using the predictor-corrector (both on a 2nd plot).

Repeat for $j_{max} = 41$ ($i_{refine} = 1$) (both on a 3rd plot) and note any comments.

Plot the pressure versus x for the shock location varying as $x_{sh} = 3: 0.5: 9$ for $j_{max} = 161$ ($i_{refine} = 4$) and using the predictor-corrector for space marching (all 7 pressure distributions for the 7 shock locations on a 4th plot). Extract the resulting exit pressures and plot the exit pressure as a function of shock location (5th plot). Any comments on how this varies? Is this what you expect based on your analysis from ENAE311?

D) Shock Capturing for Steady Flow

Assume these steady problems with shock capturing that initialize with $x_{sh} = 3.125$ which should give exit pressure of $1/\gamma$ (atmospheric). To keep the flow steady towards the end you can set variation of exit pressure to zero ($dpvar = 0.0$).

Plot pressure vs. x for an exit pressure of $1/\gamma$ (atmospheric) w/ $j_{max} = 81$ points ($i_{refine} = 2$) calculated using the van Leer flux splitting and compare to that from space marching using the predictor-corrector (both on a 6th plot). This is what is calculated before the

unsteady pressure is applied at the exit. Repeat w/ $j_{\max} = 161$ points ($i_{\text{refine}} = 4$) (both on a 7th plot). Where is the shock located? How is the shock capturing solution different from the shock marching solution?

E) Unsteady Flow

Assume for all of these unsteady problems that $x_{\text{sh}} = 3.125$.

Plot the envelope of pressure versus x for a sinusoidal variation of the exit pressure of 10% ($dp_{\text{var}} = 0.10$) for $j_{\max} = 161$ ($i_{\text{refine}} = 4$) for a moderate time period ($\text{timeper} = 1$) for the final period of six periods ($i_{\text{per}} = 6$) (8th plot). In addition plot the pressure versus time just *downstream* of the predicted shock location (at $x = x_{\text{sh}} + 0.05 \cdot x_{\text{length}}$) for the six periods (9th plot). (Don't forget to label properly!) Comments?

Repeat except for a longer time period ($\text{timeper} = 2$) (10th and 11th plots). Comments?

Repeat except for a shorter time period ($\text{timeper} = 1/2$) (12th and 13th plots). Comments?

Repeat for the moderate time period ($\text{timeper} = 1$) except for a smaller exit pressure variation of only 2% ($dp_{\text{var}} = 0.02$) (14th and 15th plots). What happens if one uses the moderate time period ($\text{timeper} = 1$) with a very large exit pressure variation ($dp_{\text{var}} = 0.20$)? Include plots if needed (16th and 17th plots). What is happening? Comments?

For one of the unsteady problems plot the Mach number versus x (18th plot). Does this tell you anything else, other than what you learned from the pressure versus x ?

F) Additional Thoughts

Discuss the solutions. How could you verify the accuracy of the space marching methods? How could you verify the unsteady solutions? Were the chosen parameters sufficient for understanding steady and unsteady nozzle flow? What did you learn from this numerical exercise? What do you think are the advantages/ disadvantages of using MATLAB versus python or C? How would you suggest making this numerical experiment better in the future?

Obviously, follow all of the requirements for a lab report as given by Dr. Silbaugh for this class.